

Selected Boundary-Layer Theorem

This note implements the hybrid route suggested by the Plan 1 and Plan 2 agent reports. The guiding point is that the pure level-set method loses control at the boundary for arbitrary rough domains, while the Plan 1 selected minimizers have exactly the regularity needed to take small positive torsion levels as controlled parallel surfaces.

The result below is split into two parts: an unconditional near-boundary good-level theorem, using only the exact level-set identity and the profile-gap estimate; and a selected-minimizer upgrade whose Serrin input is reduced in `weighted-serrin-collar-reduction.md`.

Throughout,

$$\begin{aligned} -\Delta u &= 1 \quad \text{in } U, & u &= 0 \quad \text{on } \partial U, \\ E_t &= \{u > t\}, & m(t) &= |E_t|, & L &= |U|, & \delta_T(U) &= E(U) - E(B_L), \end{aligned}$$

where B_L is the ball of volume L . Write

$$c_n = n^2 \omega_n^{2/n}.$$

For regular levels define

$$D_H(t) = \left(\int_{\{u=t\}} \frac{1}{|\nabla u|} \right) \left(\int_{\{u=t\}} |\nabla u| \right) - P(E_t)^2$$

and

$$D_I(t) = P(E_t)^2 - c_n m(t)^{2-2/n}.$$

The exact identity from `level-set-deficit-identity.md` is

$$2\delta_T(U) = \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{c_n m(t)^{1-2/n}} dt.$$

1 Near-Boundary Good Levels

Let

$$R_L = \left(\frac{L}{\omega_n} \right)^{1/n}, \quad b(s) = u_{B_L}^*(s) = \frac{R_L^2 - (s/\omega_n)^{2/n}}{2n}.$$

For $0 < \eta < L/4$, set

$$S_\eta = [L - 2\eta, L - \eta], \quad T_\eta = u^*(S_\eta).$$

Equivalently, $t \in T_\eta$ means

$$L - 2\eta \leq m(t) \leq L - \eta.$$

Proposition 1.1: boundary-slab good level. There are dimensional constants $c, C > 0$ such that if

$$0 < \eta < L/4, \quad \delta_T(U) \leq cR_L^2\eta,$$

then T_η contains a regular value t_η satisfying

$$\eta \leq |U \setminus E_{t_\eta}| \leq 2\eta$$

and

$$D_H(t_\eta) + D_I(t_\eta) \leq C c_n L^{1-2/n} \frac{\delta_T(U)}{R_L^2 \eta / L}.$$

Equivalently, since $R_L^2 \simeq_n L^{2/n}$,

$$D_H(t_\eta) + D_I(t_\eta) \leq C_n \frac{L^{2-4/n}}{\eta} \delta_T(U).$$

Moreover the chosen level lies at the correct boundary height:

$$c_n \frac{R_L^2}{L} \eta \leq t_\eta \leq C_n \frac{R_L^2}{L} \eta.$$

Proof. Let $v(s) = u^*(s)$ and $h(s) = b(s) - v(s)$. The profile-gap estimate gives

$$0 \leq h(s) \leq \frac{2\delta_T(U)}{s}.$$

On S_η , $s \geq L/2$, hence

$$0 \leq h(s) \leq \frac{4\delta_T(U)}{L}.$$

For $s \in S_\eta$, elementary bounds for $(1 - \rho)^{2/n}$, $0 \leq \rho \leq 1/2$, give

$$c_n \frac{R_L^2}{L} \eta \leq b(s) \leq C_n \frac{R_L^2}{L} \eta.$$

After decreasing c , the hypothesis $\delta_T(U) \leq cR_L^2\eta$ makes $v(s) = b(s) - h(s)$ comparable to $R_L^2\eta/L$ throughout the slab.

The same estimate gives a lower bound for the length of T_η . Indeed,

$$|T_\eta| = v(L - 2\eta) - v(L - \eta)$$

and

$$b(L - 2\eta) - b(L - \eta) \geq c_n \frac{R_L^2}{L} \eta.$$

The oscillation of h across the slab is at most $4\delta_T(U)/L$, so, again by the smallness assumption,

$$|T_\eta| \geq c_n \frac{R_L^2}{L} \eta.$$

Now average the exact identity over T_η . Since $m(t) \in [L - 2\eta, L - \eta] \subset [L/2, L]$ on T_η , there exists a regular $t_\eta \in T_\eta$ with

$$D_H(t_\eta) + D_I(t_\eta) \leq \frac{2\delta_T(U)}{\int_{T_\eta} \frac{dt}{c_n m(t)^{1-2/n}}} \leq C_n \frac{L^{2-4/n}}{\eta} \delta_T(U).$$

The volume-layer bound follows directly from the definition of T_η .

2 Transfer Back to the Original Set

The selected level is useful only if stability of E_{t_η} can be returned to U .

Lemma 2.1: boundary-layer asymmetry transfer. For every $t > 0$, with $\eta = L - m(t)$,

$$\mathcal{A}(U) \leq \mathcal{A}(E_t) + 2\frac{\eta}{L}.$$

Proof. Let B_s be an optimal ball for E_t , where $s = m(t) = L - \eta$, and let B_L be the concentric dilation of B_s with volume L . Since $E_t \subset U$,

$$|U \Delta B_L| \leq |U \setminus E_t| + |E_t \Delta B_s| + |B_L \setminus B_s|.$$

Divide by L . The first and third terms are both η/L , while the middle term is $(s/L)\mathcal{A}(E_t) \leq \mathcal{A}(E_t)$.

Corollary 2.2: good-level transfer estimate. If t_η is the level from Proposition 1.1, then

$$\mathcal{A}(U)^2 \leq 2\mathcal{A}(E_{t_\eta})^2 + 8\frac{\eta^2}{L^2}.$$

By the quantitative isoperimetric inequality applied to E_{t_η} ,

$$\mathcal{A}(E_{t_\eta})^2 \leq C_n \frac{D_I(t_\eta)}{m(t_\eta)^{2-2/n}},$$

and therefore

$$\mathcal{A}(U)^2 \leq C_n \frac{D_I(t_\eta)}{L^{2-2/n}} + 8\frac{\eta^2}{L^2}.$$

Choosing

$$\eta = L \delta_{\text{rel}}^{1/2}, \quad \delta_{\text{rel}} := \frac{\delta_T(U)}{L^{1+2/n}},$$

one obtains the scale-invariant estimate

$$\mathcal{A}(U)^2 \leq C_n \delta_{\text{rel}}^{1/2}$$

from level-set information alone. This is the known boundary-layer loss: it gives the wrong exponent. The selected-minimizer regularity is needed to replace this average estimate by a boundary trace estimate.

3 Selected-Minimizer Boundary Trace

Assume now that $U = \{u > 0\}$ is a Plan 1 selected minimizer in the graph regime. Thus ∂U is $C^{1,\gamma}$, the Bernoulli coefficient

$$q(y) = |\nabla u(y)|, \quad y \in \partial U,$$

satisfies

$$0 < q_- \leq q \leq q_+, \quad [q]_{C^\gamma(\partial U)} \leq M,$$

and the interior normal exponential map is a $C^{1,\gamma}$ diffeomorphism in a fixed collar of ∂U .

Lemma 3.1: positive levels are controlled parallel surfaces. There are $t_0 > 0$ and C , depending only on the selected-minimizer regularity constants, such that for $0 < t < t_0$

$$\{u = t\} = \left\{ y - \frac{t}{q(y)} \nu_U(y) + R_t(y) : y \in \partial U \right\},$$

with

$$|R_t(y)| \leq Ct^{1+\gamma}.$$

Consequently,

$$|U \setminus E_t| = t \int_{\partial U} \frac{1}{q} d\mathcal{H}^{n-1} + O(t^{1+\gamma}),$$

$$P(E_t) = P(U) + O(t),$$

and

$$|\mathcal{A}(E_t) - \mathcal{A}(U)| \leq Ct.$$

Proof. In the collar, write points as $x = y - r\nu_U(y)$. Boundary $C^{1,\gamma}$ regularity and the Bernoulli condition give

$$u(y - r\nu_U(y)) = q(y)r + O(r^{1+\gamma})$$

uniformly in y . Since $q \geq q_- > 0$, the equation $u = t$ has a unique solution

$$r_t(y) = \frac{t}{q(y)} + O(t^{1+\gamma})$$

by the elementary inverse estimate for $C^{1,\gamma}$ perturbations of a linear function. The area and volume expansions follow from the area formula for the normal map. The asymmetry estimate follows from $|E_t \Delta U| = |U \setminus E_t| = O(t)$ and the fixed-volume Lipschitz continuity of Fraenkel asymmetry under symmetric difference, after the harmless volume normalization.

4 The Conditional Sharp Theorem

The preceding lemmas isolate the only missing analytic input.

Input S: weighted Serrin stability on selected collars. For selected minimizers in the graph regime, assume the following stability estimate holds uniformly for all sufficiently small regular levels:

$$\mathcal{A}(E_t)^2 \leq C \left(\frac{D_H(t) + D_I(t)}{L^{2-2/n}} + \frac{|U \setminus E_t|^2}{L^2} \right).$$

The D_I part is the quantitative isoperimetric inequality. The real content is to upgrade the weighted variance identity

$$\int_{\{u=t\}} \frac{(|\nabla u| - \bar{f}_t)^2}{|\nabla u|} = \frac{m(t)}{P(E_t)^2} D_H(t)$$

to the boundary norm required by a Serrin-stability theorem, using the uniform C^γ estimates supplied by Plan 1.

The follow-up note `weighted-serrin-collar-reduction.md` observes that this specific Input S is already implied by the quantitative isoperimetric inequality through $D_I(t)$. The useful role of $D_H(t)$ is diagnostic: it provides approximate Serrin data, but it is not the sharp Fraenkel bottleneck in the selected graph regime.

Theorem 4.1: selected boundary-layer theorem, conditional on Input S. Let U be a Plan 1 selected minimizer in the graph regime, with $|U| = L$, and let

$$\delta_{\text{rel}} := \frac{\delta_T(U)}{L^{1+2/n}}.$$

Assume Input S. If δ_{rel} is sufficiently small, then there is a regular level t such that

$$|U \setminus E_t| \leq CL\delta_{\text{rel}}^{1/2},$$

E_t is a regular approximate Serrin domain,

and

$$\mathcal{A}(U)^2 \leq C \left(\frac{D_H(t) + D_I(t)}{L^{2-2/n}} + \delta_{\text{rel}} \right).$$

In particular,

$$\mathcal{A}(U)^2 \leq C\delta_{\text{rel}}^{1/2}$$

unconditionally from the averaged level-set bound, and the sharp estimate

$$\mathcal{A}(U)^2 \leq C\delta_{\text{rel}}$$

would follow once the boundary deficit propagation step finds a boundary level with

$$D_H(t) + D_I(t) \leq CL^{2-2/n}\delta_{\text{rel}}.$$

Proof. Choose $\eta = L\delta_{\text{rel}}^{1/2}$. Proposition 1.1 gives a regular t with $|U \setminus E_t| \leq 2\eta$ and controlled level deficit. Lemma 3.1 ensures that, for small δ_{rel} , this level is a regular parallel surface in the selected collar. Input S gives the displayed estimate for $\mathcal{A}(E_t)^2$, and Lemma 2.1 transfers it back to U . The boundary-layer volume contributes exactly $\eta^2/L^2 = \delta_{\text{rel}}$.

The last statement identifies the remaining sharpness gap: the current averaging argument gives a $1/\eta$ loss, while the selected-minimizer trace theorem should use the small-collar regularity to avoid averaging over a slab whose length is only $O(\eta)$.

5 Implementation Consequence for the Full Strategy

This note makes the Plan 1/Plan 2 hybrid route precise:

1. Plan 1 selects U and preserves the quotient $(E(\Omega) - E(B))/\alpha(\Omega)$.
2. Plan 1 graph entry gives a $C^{1,\gamma}$ collar and Bernoulli bounds.
3. Plan 2 gives the exact integrated level-set identity.
4. Proposition 1.1 finds a genuine near-boundary level.
5. Lemma 3.1 turns that level into a controlled trace of ∂U .
6. The remaining non-formal input is boundary deficit propagation: $D_I(0) + D_H(0) \lesssim E(U) - E(B)$, or an equivalent Plan 1 near-spherical expansion.

Thus the most promising next concrete task is the Plan 1 graph-entry and near-spherical closure, not another selection lemma or a stronger Serrin theorem for rough levels.